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iNaturalist provides valuable data for monitoring non-native herpetofauna

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Conflict of interest

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iNaturalist provides valuable data for monitoring non-native herpetofauna

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Abstract

Introduced herpetofauna are an increasing conservation concern, yet monitoring their range and spread remains challenging due to the diversity of taxa and available resources. We assessed two prominent species occurrence reporting platforms, iNaturalist and EDDMapS, for tracking the distribution of introduced reptiles and amphibians across Florida. We compared the availability of data across both platforms while assessing the spatial coverage of species using minimum convex polygons. iNaturalist yielded a higher total number of introduced herpetofauna observations across a broader geographic range, particularly in urban areas, whereas EDDMapS reported greater species richness, likely due to its focus on rapid response. While total observations were strongly associated with minimum convex polygons, the platform type had no significant effect on range size when accounting for effort. Our results demonstrate the value of participatory citizen science for early detection and long-term monitoring of introduced herpetofauna, suggesting that a centralized, widely used platform like iNaturalist may offer increased efficacy for statewide monitoring efforts.

Keywords: Amphibian, Citizen Science, Data monitoring, Detection, Florida, Invasive, Reptile, Urbanization

Introduction

With an increasingly connected world and growing human population, biological invasions have become an important focus of conservation and management ecology [1,2]. Within the last 50 years, research has examined the extent of anthropogenic alterations to ecosystems that have facilitated the intentional or accidental spread, and possible establishment of non-native plant and animal species [2,3,4]. While not all introduced species are classified as invasive and potentially harmful to an ecosystem, all species that are classified as invasive are non-native to their newly established range [5,6]. The establishment of species outside of their native range is uncommon; however, the species that become invasive causes severe negative ecological impacts on the ecosystem [7,8]. Once these invasive species are established, they can outcompete natives for space and resources [9] as well as shift trophic levels, directly impacting 58% of species on the Endangered Species Act [10,11]. Once a species becomes established, it becomes increasingly challenging to eradicate [12,13]. This emphasizes the importance of early detection and rapid response of introduced species [14,15]. By identifying and tracking emerging populations before they reach the point of ecological dominance, natural resource managers can mitigate damage more effectively. This is especially true for taxa such as reptiles and amphibians, which are often cryptic, understudied, and are able to persist for a long period of time without detection [16,17].

Herpetofauna or “herps” represent the classes of Reptilia and Amphibia, which are frequently studied together due to their similar ecology, physiology, and historical progression [18]. Herpetofauna have historically been understudied in wildlife research, as funding has often favored more charismatic vertebrates [19,20]. However, due to their large-scale declines in the last half a century, interest and research have increased, with more studies focusing on herp

conservation, management, and distribution [21,22]. Herpetofauna are common non-native species in subtropical climates due to their popularity as pets, a lack of natural predators, and their cold-blooded physiology [23,24]. In these conditions, non-native herps have a higher establishment rate, allowing them to outcompete and potentially displace native herps [24,25]. Florida currently has one of the largest populations of non-native herpetofauna globally, which have led to dramatic changes in ecosystem functionality, biodiversity, and species distribution [26] (Hegan, 2014). There are around 137 taxa introductions documented from 1963 to 2010 [25,27]. Species such as *Salvator merianae* (Argentine black and white tegu), *Iguana iguana* (Green Iguana), and *Python bivittatus* (Burmese python) are among some of the most problematic non-native reptile species introduced to Florida as they outcompete native species for food while displacing them from their habitat [24,28,29]. Many of the non-native herp species in Florida are fairly cryptic and difficult to observe. These issues mainly stem from their camouflage patterns, tendency to hide from observation, and limited visibility, making them difficult to spot, resulting in an underestimation of their populations and distribution [17,30]. This cryptic nature poses challenges for effective monitoring and management, hindering timely conservation responses to non-native herps' ecological impacts.

Monitoring herp populations has traditionally relied on in-field methods including visual encounter surveys, pitfall traps with drift nets, coverboards, and mark-recapture surveys, each requiring intensive field effort and expert identification skills [31]. In recent decades, a new method of tracking herps has emerged through the use of EDDMapS (Early Detection & Distribution Mapping System). This system's primary purpose is to provide a real-time picture of invasive and pest species distribution across North America, helping to identify newly established species and their spread [32,33]. EDDMapS aggregates data from a wide range of

sources including databases, organizations, and volunteer observations attempting to best provide early warning signs to conservation biologists and land managers [34]. While EDDMapS can be a useful tool in detecting the early establishment of introduced species, it has a much smaller public audience compared to broad-scale participatory citizen science platforms. For example, the citizen science platform, iNaturalist (www.inaturalist.org), has had significant increases in user activity over recent years [35,36]. iNaturalist is an application available to the general public, where users can identify plant and animal species by submitting photos or audio recordings of individuals they observe in nature [37,38]. This platform also provides an online forum where observations are digitally vetted by other users, ensuring that the identification of plant and animal has a higher degree of accuracy [37]. With iNaturalist's extensive user base and consistent user contributions, it has become one of the most influential and reliable tools in determining ecological biodiversity within systems globally [38,39]. Consequently, the iNaturalist platform has been used in documenting introduced species as users do not need prior knowledge of species status when uploading observations [17, 40-42]. While both EDDMapS and iNaturalist may offer valuable data on introduced species, they have not been directly compared to investigate their effectiveness and potential advantages and disadvantages. Because these platforms serve different audiences, with EDDMapS primarily used by land managers and professionals, whereas iNaturalist receives a broad base of public users, a comparative analysis could provide a more comprehensive understanding of how non-native species distributions can be studied and monitored in the future.

Our aim was to compare the effectiveness of EDDMapS and iNaturalist for detecting non-native herpetofauna across Florida. Specifically, we had three objectives: 1) to comprehensively compare the data available from iNaturalist and EDDMapS for all recorded

non-native herpetofauna in Florida; 2) to understand the differences in data availability as a function of urbanization level; and 3) to understand the overall distribution (i.e., range size) of non-native species across Florida. This comparison will provide insights into how different data streams contribute to our understanding of species invasions and their spatial patterns.

Methods

iNaturalist

iNaturalist was created in 2008 as a multi-taxa citizen science platform hosted by the California Society of Sciences and National Geographic Society [43]. Gradually, this platform expanded until it became an independent nonprofit in 2023. On the platform, users submit photos and/or audio recordings of organisms, and are asked to provide the location, date and time, and an initial identification of the species with the help of Computer Vision, a machine learning algorithm that analyzes imagery to provide a list of identification suggestions. Afterwards, members of the iNaturalist community provide their identification of the organism. Observations with more than two thirds agreement on the species identification and that pass the iNaturalist Data Quality Assessment [44] acquire a “Research Grade” status, indicating that the observation has been validated and can be used for ecological research [37]. To obtain iNaturalist observations, we used the Global Biodiversity Information Facility (GBIF), which stores all “Research Grade” iNaturalist observations that are shared with the appropriately specified license. Specifically, on August 29, 2024, we obtained all Florida records of class Amphibia and Reptilia [45]. We then filtered data to include observations taken between January 2014 and August 2024. The start date was selected as it coincides with a significant increase in iNaturalist user adoption and data quality improvements [46]. We manually determined which species on iNaturalist were non-native, and filtered out all native species observations (Table S1). For the

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3 first objective, where we compared the number of species present on iNaturalist versus
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5 EDDMapS, we also conducted a manual search directly on the iNaturalist platform. In a few
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7 cases, we identified “Research Grade” observations of non-native species that were not included
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9 in the GBIF dataset. This occurred when users marked their images as “all rights reserved,”
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11 thereby preventing the data from being shared with GBIF. However, for the analyses conducted
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13 under the second and third objectives, we relied on the iNaturalist data exported via GBIF, as this
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15 is the recommended and standardized approach for accessing “Research Grade” iNaturalist
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17 observations.
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23 *EDDMapS*

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26 EDDMapS is a web-based mapping system for documenting invasive species and pest
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28 distribution (www.eddmaps.org). Since its launch in 2005, the observations have expanded to
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30 account for 8.3 million non-native organisms across North America. This platform was created
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32 as a centralized database for agencies, programs, partners, and individuals working on non-native
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34 and invasive species, while also welcoming contributions from the general public [47].
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36 Submitted observations are reviewed and verified by a network of trained local and state experts
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38 before being published on the site [48]. Data is made available to provide producers and
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40 stakeholders up-to-date information on the spread and distribution of invasive species [49].
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42 Applications such as EDDMapS can retrieve introduced species data to some degree from
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44 iNaturalist, depending on the quality and geoprivacy of individual observations [50].
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49 In Florida, EDDMapS has partnered with the Florida Fish and Wildlife Conservation
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51 Commission (FWC), Florida Natural Areas Inventory, Florida Invasive Species Partnership,
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53 Florida Exotic Pest-Plant Council, National Park Service, U.S. Fish and Wildlife Service, The
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Nature Conservancy, and the University of Florida to create a centralized database of their invasive species records [47]. Other agencies and members of the public may also submit records to the platform, which has been promoted through the IveGot1 (www.eddmaps.org/florida/) website, application, and hotline for reporting invasive animals and plants in Florida. In Florida, a majority of herpetofaunal EDDMapS data comes from state agencies (87.3% of all herp records between 2014 and 2024 with classified owners), with only 8.5% coming from the general public. While iNaturalist records are occasionally collected in EDDMaps, we found that was rare (0.22% of herpetofauna EDDMapS records between 2014 and 2024).

We requested and obtained all EDDMapS observations of introduced reptile and amphibian species across Florida from January 2014 to August 2024. Individual species observations extracted from EDDMapS were confirmed to be non-native, with those classified as native to Florida removed from the data.

Data aggregation

All data collected from both iNaturalist and EDDMapS were downloaded and cleaned using R version 4.4.3 [51], relying on the Tidyverse package for data manipulation [52]. Data from both iNaturalist and EDDMapS were assessed to ensure that all species included were introduced to Florida. All species names were converted to traditional binomial nomenclature to ensure continuity between iNaturalist and EDDMapS data (Table S1).

Platform Data Availability Comparison

To assess the data availability between iNaturalist and EDDMapS for all non-native herps in Florida, aggregated data were joined by species name and identified based on the application. Total observations of each introduced herp species across each platform were reported as

numeric values across all years. The R package *ggplot2* was used to generate maps and visual relationships between herp species across iNaturalist and EDDMapS [53]. Maps were generated for both iNaturalist and EDDMapS data to indicate the total number of introduced herp species per Florida county. We assessed the relationship between the number of observations of each species from iNaturalist and EDDMapS by fitting negative binomial generalized linear model — to account for overdispersion in the observation data — using the number of iNaturalist observations per species as the predictor variable and the number of EDDMapS observations as the response variable [54].

Data availability as a function of urbanization level

To understand the data availability in both iNaturalist and EDDMapS across an urbanization gradient, Florida population data was extracted from Google Earth Engine as a Tagged Image File (.tif) containing the estimated number of persons per square kilometer [55]. The R package *raster* was used to extract population values for observations in the iNaturalist and EDDMapS datasets [56]. Urbanization level was kept as a continuous variable across both iNaturalist and EDDMapS observations. We used a generalized linear model with a gaussian distribution and identity link to test whether the log-transformed population density differed between iNaturalist and EDDMapS. Data showed a positive skew, hence why it was log-transformed for the generalized linear model. The model included data source as a categorical predictor to assess if the source of observations was influenced by the response variable, population density.

Range distribution

To understand the distribution of non-native individuals from both EDDMapS and iNaturalist across Florida, minimum convex polygons (MCPs) were generated for species with more than 5 observations on both iNaturalist and EDDMapS. MCPs were fit using the *adehabitatHR* package in R [57]. MCPs were generated to determine the range of introduced herp species observed both in iNaturalist and EDDMapS. For home range analysis, MCPs were selected instead of kernel density estimations (KDE) due to their ability to generate straightforward, standardized polygons that are more practical for management purposes, in contrast to KDE methods where bandwidth and smoothing parameters can affect home range outcomes [58,59]. All MCPs were set to 95% to estimate the primary occurrence of all recorded species, with 5% representing possible outliers and GPS errors in observations [60,61].

Generalized linear models were fitted to examine the differences in log-transformed MCP areas between iNaturalist and EDDMapS, while controlling for the log-transformed number of observations, as both variables were positively skewed. The R package *lme4* was then used to conduct mixed effect models with species as a random effect variable [62] because we were interested in the overall difference between both platforms (i.e., average effect size), averaged across different species.

Data Availability

All data and code used in this study are available through a GitHub repository:
<https://github.com/MarioZuliani/iNaturalist-vs-EDDMapS-Herpetofauna>.

Results

Platform Data Availability Comparison

A total of 141,360 observations were collected from both EDDMapS ($n = 57,222$) and iNaturalist ($n = 84,138$) from 2014 to 2024. Across both datasets 151 herp species were observed, with 18 unique species from iNaturalist and 62 unique species from EDDMapS. A total of 71 species were observed in both iNaturalist and EddMapS (Supplement Table S1). The top 5 most commonly observed introduced herp species in iNaturalist were brown anole (*Anolis sagrei*; $N = 37,838$), Cuban tree frog (*Osteopilus septentrionalis*; $N = 9,123$), green iguana (*Iguana iguana*; $N = 7,585$), northern curly-tailed lizard (*Leiocephalus carinatus*; $N = 5,322$), and Peter's rock agama (*Agama picticauda*; $N = 3,761$). A total of 21 species were reported only once on iNaturalist, while only 9 species were reported once on EDDMapS. The top 5 most commonly observed introduced herp species in EDDMapS included green iguana (*Iguana iguana*; $N = 10,252$), Burmese python (*Python bivittatus*; $N = 9,899$), Argentine black and white tegu (*Salvator merianae*; $N = 9,043$), Peter's rock agama (*Agama picticauda*; $N = 5,058$), and brown anole (*Anolis sagrei*; $N = 4,512$). Spatially across Florida, the county comparison showed that Miami-Dade County had the most observations for iNaturalist ($n = 24,714$) and EDDMapS ($n = 24,209$) (Figure 1A). All counties contained at least 1 iNaturalist observation of introduced herpetofauna with Madison County and Lafayette County having the lowest number of observations ($n = 2$; Figure 1A). Liberty, Hamilton, and Holmes counties contained no introduced herpetofauna data from EDDMapS ($n = 0$; Figure 1A).

There was a significant positive association between the number of observations recorded in iNaturalist and those in EDDMapS by species ($\beta = 0.59$, $SE = 0.08$, $z = 7.66$, $p < 0.001$, Figure 1B). We found that, for every 1% increase in iNaturalist observations, EDDMapS observations increase by 0.59%. The model's dispersion parameter ($\theta = 0.38$, $SE = 0.04$) indicated substantial overdispersion in the data.

Data availability as a function of urbanization level

The mean population density for EDDMapS observations was 536 ± 4 people per km^2 , while iNaturalist observations occurred in areas with a mean population density of approximately 656 ± 2 people per km^2 . Observations between EDDMapS and iNaturalist significantly differed across an urbanization gradient ($t_{(296337)} = 46.45$, $p < 0.001$) with EDDMapS observations tending to be in less urban areas than iNaturalist observations ($\beta = 0.608$, $\text{SE} = 0.013$, $t = 46.45$, $p < 0.001$; Figure 2). This coefficient indicates that iNaturalist observations occurred in areas with population densities approximately 22% higher than those recorded in EDDMapS.

Range distribution

The spatial extent of species reporting, as measured by MCP area, was strongly associated with the total number of observations ($\beta = 0.55$, $\text{SE} = 0.11$, $t = 5.01$, $p < 0.001$; Supplement Figure S1), with each 10% increase in observations corresponding to a 6% larger MCP area. When controlling for observation counts, there was no significant difference in MCP area between EDDMapS and iNaturalist ($\beta = 0.31$, $\text{SE} = 0.23$, $t = 1.38$, $p = 0.17$). The random effect of species tested in the mixed effect model accounted for substantial variation in range size ($\sigma^2 = 0.38$; Figure 3), suggesting inherent differences in spatial extent among species is not explained by observation count or platform.

Discussion

Our findings demonstrate that both iNaturalist and EDDMapS provide extensive data on introduced herpetofauna in Florida but they differ in coverage and spatial bias, particularly in

terms of the urbanization gradient they sample. Both platforms contributed extensive data with 151 total introduced herpetofauna species reported, but only 71 species were shared between applications, highlighting important differences in species coverage. iNaturalist recorded more total observations than EDDMapS, yet the spatial overlap and the significant positive association between species observations across platforms suggest some degree of complementarity. There was no significant difference in range size between the platforms when accounting for observation count, suggesting that both offer comparable information on species distributions given they contain the same quantity of data. Together, our findings support the notion that integrating data from both platforms provides a more comprehensive understanding of introduced herpetofauna distributions.

Platform Data Availability Comparison

Our first objective aimed to comprehensively compare the data available from iNaturalist and EDDMapS for all recorded introduced herpetofauna species in Florida. The comparison between both platforms highlights key differences in data availability and how each platform captures herpetofauna observations in Florida, reflecting their distinct user bases and data collection goals. The greater number of total observations in iNaturalist likely stems from its accessibility and popularity among casual users and citizen scientists [35,37]. iNaturalist's mobile-friendly interface, real-time feedback, and social features encourages more frequent reporting, primarily in densely populated urban areas where more people encounter and report wildlife [39]. This user-friendly system makes reporting introduced herpetofauna easier for non-specialized observers [37,39]. In contrast, EDDMapS is oriented more towards the early detection and rapid response by professionals or volunteers who are aware of non-native species to report through IveGot1, which may explain its higher species richness despite having fewer

observations [33]. Users contributing to EDDMapS are typically more taxonomically informed or may be targeting specific hotspots of invasions, leading to more comprehensive species documentation despite the lower volume [31]. With only 71 species overlapping between iNaturalist and EDDMapS, this suggests that there are meaningful differences in data selectivity and observer expertise between both platforms. For example, species such as the Argentine black and white tegu, while common in EDDMapS observations, was relatively underrepresented in iNaturalist, possibly due to trapping programs for this species [63], which increases the rate of professional reporting compared to iNaturalists users who are subjected to their less conspicuous behavior or low detectability in areas where observers commonly frequent [64]. The dominance of brown anoles and green iguanas reflects the bias towards urban areas by both platforms and the population density of these species across their range [23,65,66].

Geographically, the near-universal county coverage in iNaturalist compared to notable gaps in EDDMapS underscores the utility of crowdsourced platforms to broaden the spatial distribution of sampling of introduced species. These differences are likely linked to broader spatial distribution of iNaturalist users particularly in areas where professional monitoring may be somewhat limited, such as on private properties [35,37]. The modest correlation between platforms' species observation counts reflects both shared patterns (common species remaining common) and distinct differences stemming from user behavior, observer effort, and platform intent. Ultimately, our findings suggest that iNaturalist excels at providing large volumes of data on established herpetofauna, while EDDMapS can capture a wider range of species. However, the high proportion of species reported less than 5 times on EDDMapS suggests that this platform is better able to capture rarer species or early detection events that may be missed by citizen science platforms.

Data availability as a function of urbanization level

Urbanization is a key factor shaping both biological invasions and data collection, as it influences where people live and their engagement with wildlife [67]. Our results revealed that iNaturalist observations were significantly more concentrated in densely populated areas compared to EDDMapS. This disparity likely reflects iNaturalist's reliance on casual contributors who are more active in urban and suburban settings, aligning with known biased sampling in participatory science platforms [68-70]. In contrast, EDDMapS observations were more evenly distributed, with some bias towards less populated areas, which is consistent with its use by professionals who may be conducting more targeted monitoring in rural or natural landscapes [48]. These results reinforce the notion that each platform captures a distinct subset of introduced herpetofauna; iNaturalist through opportunistic, urban-driven engagement, and EDDMapS through structured, and often management-oriented surveying. Recognizing this complementary dynamic is essential in maximizing the utility of both platforms in statewide biodiversity monitoring and invasive species management.

Range distribution

We found that spatial extent of species reporting was strongly associated with the number of observations, as species with more observations had significantly larger reported ranges. This highlights a common challenge in biodiversity data collection, where spatial coverage often reflects where people are looking, rather than where species are truly present [71,72]. However, the broader spatial reach and volume of citizen science observations help overcome these biases, especially when aggregated over time and across users, making platforms like iNaturalist valuable tools for detecting introduced species [65,73-75]. When controlling for the number of

observations, we found that there were no significant differences between iNaturalist and EDDMapS MCPs, suggesting that both platforms can provide comparable spatial information, given they have the same volume of data. However, the positive coefficient, though not statistically significant, for EDDMapS suggests a trend towards slightly broader spatial coverage, possibly reflecting the platforms more systematic surveying efforts or targeted reporting by trained users [76-78].

Beyond observation effort and platform, species identity explained additional variation in range size, likely reflecting urban area bias and differential detection probabilities across both platforms. Species traits such as urban affinity and conspicuousness make some species more likely to be observed by iNaturalist compared with EDDMapS. For introduced herpetofauna, these detection biases may favor species that are well-adapted to urban environments and human-modified landscapes [79,80]. For example, species such as the brown anole and green iguana are not only tolerant of urban environments but are also visible and active during daylight hours, making them more likely to be observed on both iNaturalist and EDDMapS [81]. In contrast, species with cryptic behaviors, nocturnal activity patterns, or those that avoid urban areas may be systematically underrepresented on both platforms, regardless of their true abundance and distribution [81,82]. We suggest that understanding these species-specific differences is an avenue of further research which can be helpful to interpret current observation patterns when assessing the potential spread of introduced herpetofauna in Florida’s changing landscapes.

Implications

Given that early detection and rapid response is the most cost-effective strategy for managing non-native species [84], our key finding is that both platforms can contribute valuable

data to better guide rapid response efforts. For instance, iNaturalist documented species such as the Central American Boa (*Boa imperator*) and the red-bellied short-necked slider (*Emydura subglobosa*), which were not reported on EDDMapS. Conversely, EDDMapS documented the red tegu (*Salvator rufescens*) and several monitor (*Varanus*) species absent from iNaturalist.

While this study is the first to directly compare species records across iNaturalist and EDDMapS, we did not assess what factors may influence the likelihood of a species being reported on one platform versus the other. Nor did we evaluate the spatiotemporal dynamics of new records—such as which platform was first to document new county-level occurrences of non-native species. Future research should explore these questions to further clarify the potential of both platforms in supporting early detection and rapid response initiatives.

While having multiple platforms can provide complementary insight in monitoring and detecting introduced species, the redundancy and fragmentation in data can create challenges for large-scale synthesis and response coordination [85]. Our results suggest that streamlining introduced species observations through a single, well-supported platform may improve detection efficiency, facilitate data standardization, and encourage broader participation. One way this can be achieved is through integrating EDDMapS data into the GBIF so managers can acquire both iNaturalist and EDDMapS data simultaneously [50]. iNaturalist's broad geographic coverage, high observation volume, and public accessibility make it a powerful tool for monitoring established introduced species, especially in areas not routinely assessed by professionals [37,40]. While EDDMapS provide structured data, high taxonomic reliability, and excels at early-detection of rare species, our results demonstrate that iNaturalist alone can effectively capture the distribution patterns of established introduced herpetofauna. This highlights the efficiency and scalability of leveraging widely-used citizen science platforms for

invasion monitoring, particularly when integrated with professional databases like EDDMapS for comprehensive management strategies.

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Figures

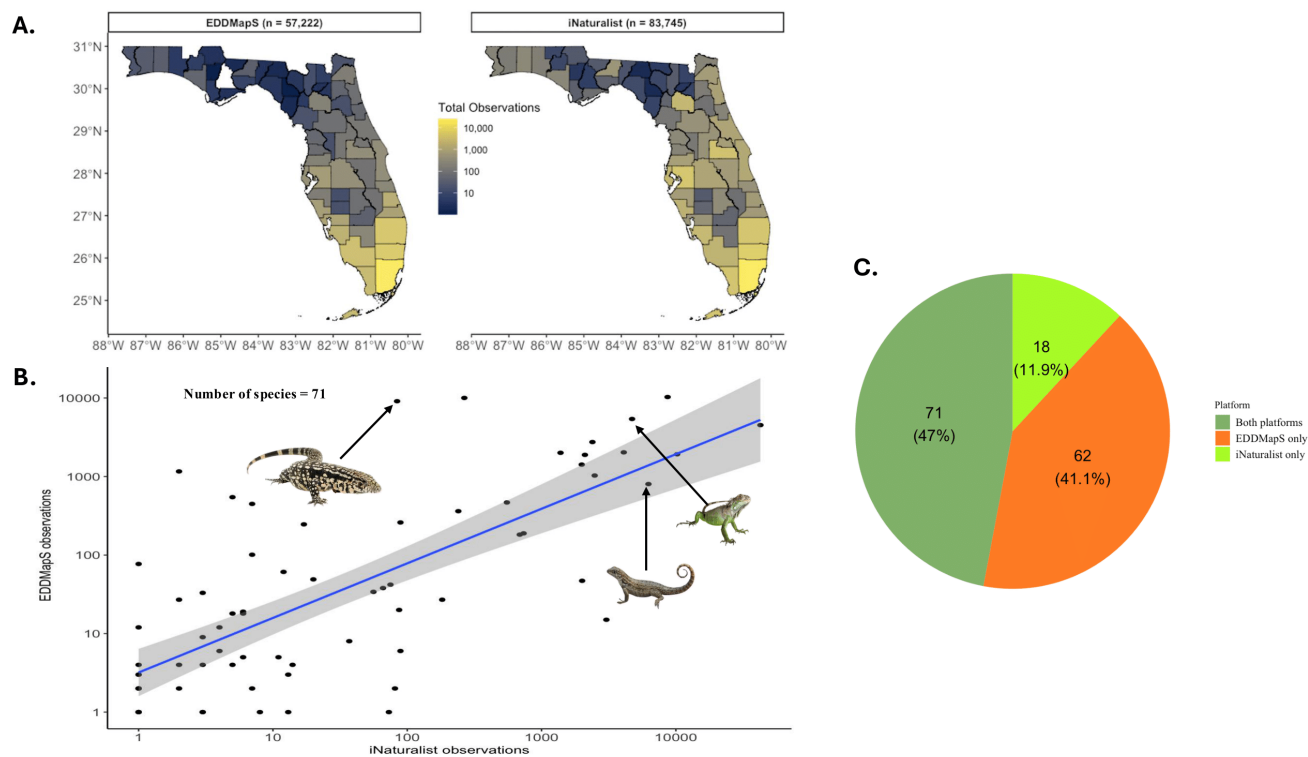


Figure 1: Comparison of introduced reptile and amphibian species across Florida from EDDMapS and iNaturalist observations. **A)** The spatial distribution of total observations per county from EDDMapS (left; n = 57,222) and iNaturalist (right; n = 84,138). Warmer colors indicate counties with more observations. **B)** Log-log scale comparison of total species-level observations between EDDMapS and iNaturalist for 71 common introduced species. The blue line represents a linear regression fit ($\pm$ 95% shaded CI) between the number of observations from EDDMapS and iNaturalist. Selected species (From Right to Left: *Iguana iguana*, *Leiocephalus carinatus*, and *Salvator merianae*) are indicated for visual reference. **C)** Comparison of non-native herpetofauna species presence between iNaturalist only (bright green), EDDMapS (orange), and both platforms (dark green) in Florida.

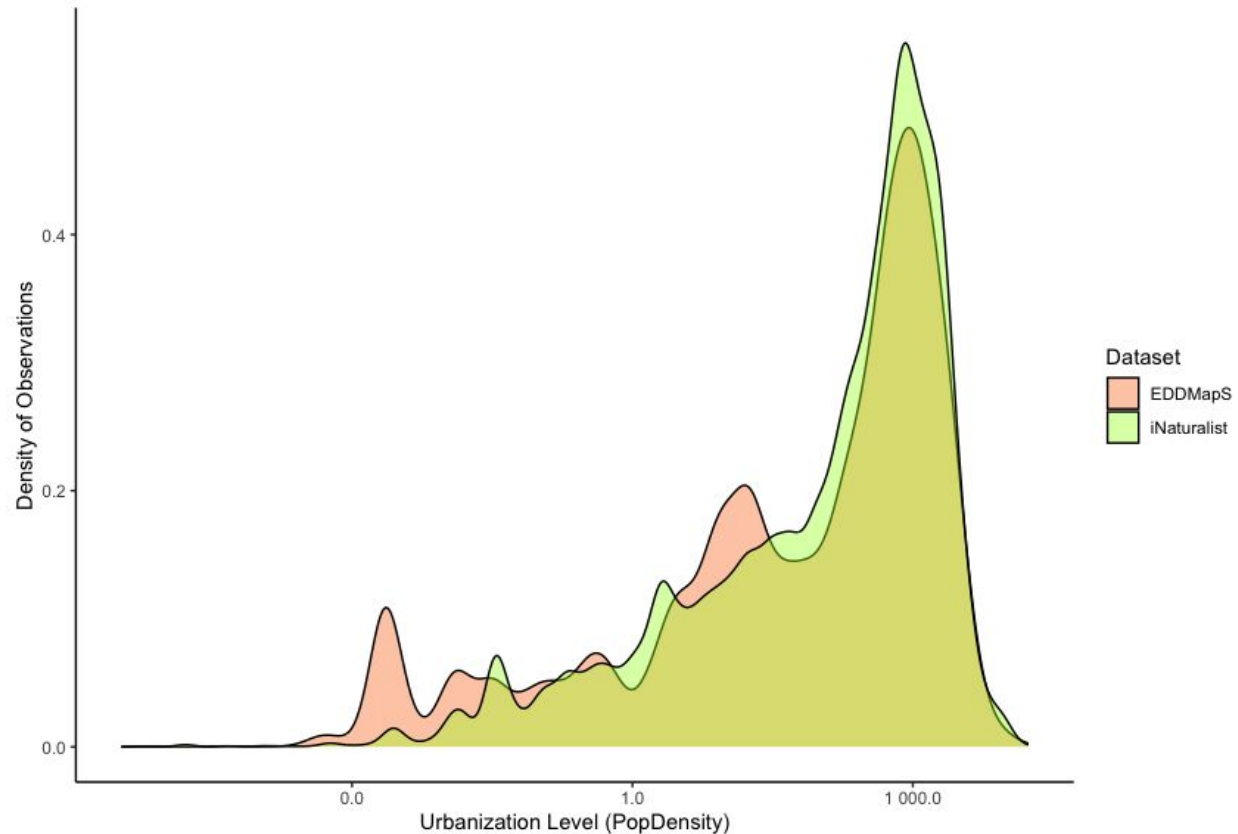


Figure 2: Distribution of introduced reptile and amphibian observations across urbanization gradients in Florida. Density plot comparing the frequency of observations from EDDMapS (orange) and iNaturalist (green) along a log-transformed urbanization gradient, represented by human population density (PopDensity). iNaturalist observations are more strongly associated with higher density areas (urbanized areas), while EDDMapS observations are more broadly distributed across low- to mid- density areas, reflecting the differences in observer behavior and data collection strategies between platforms.

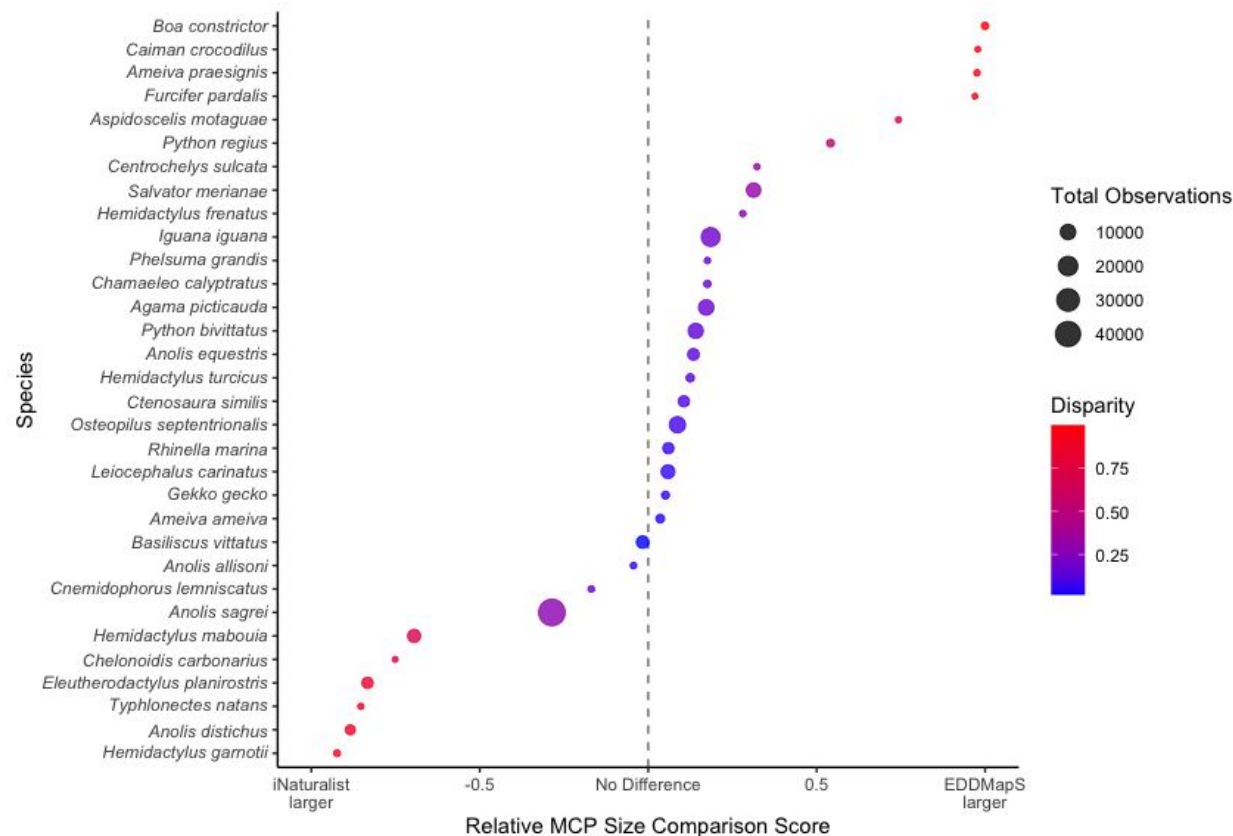


Figure 3: Comparison of species' 95% minimum convex polygon (MCP) range sizes derived from iNaturalist and EDDMapS data. Each point represents an introduced reptile or amphibian species in Florida, with the x axis showing the relative difference in relative MCP sizes between datasets. Negative values indicate larger iNaturalist ranges, while positive values indicate larger EDDMapS ranges. Point size is proportional to the total number of observations across both datasets, while the point color indicates the magnitude of disparity in MCP estimates between EDDMapS and iNaturalist. Most species display moderate to no differences, however some exhibit substantial disparity, highlighting data-specific variation in spatial coverage.

Supplementary Material:

Table S1: Presence of introduced herpetofauna species in Florida across iNaturalist and EDDMapS. Each row represents a unique species with columns indicating if the species was observed in iNaturalist or EDDMapS. A total of 151 unique herpetofauna species were reported across both platforms with iNaturalist having 18 unique individuals and EDDMapS having 62. A total of 71 herpetofauna species were common between both platforms. All species classified as research grade on iNaturalist but were excluded from GBIF are indicated with an asterisk.

<i>Species</i>	<i>iNaturalist_Present</i>	<i>EDDMapS_Present</i>
<i>Acrochordus javanicus</i>	No	Yes
<i>Afrixalus fornasini</i>	No	Yes
<i>Agama picticauda</i>	Yes	Yes
<i>Ahaetulla nasuta</i>	No	Yes
<i>Aldabrachelys gigantea</i>	No	Yes
<i>Ameiva ameiva</i>	Yes	Yes
<i>Ameiva praesignis</i>	Yes	Yes
<i>Anolis allisoni</i>	Yes	Yes
<i>Anolis chlorocyanus</i>	Yes	No
<i>Anolis cristatellus</i>	Yes	Yes
<i>Anolis cybotes</i>	Yes	No
<i>Anolis distichus</i>	Yes	Yes
<i>Anolis equestris</i>	Yes	Yes
<i>Anolis garmani</i>	Yes	No
<i>Anolis porcatus</i>	No	Yes
<i>Anolis sagrei</i>	Yes	Yes
<i>Antaresia maculosa</i>	No	Yes
<i>Aspidoscelis motaguae</i>	Yes	Yes
<i>Basiliscus plumifrons</i>	Yes	Yes
<i>Basiliscus vittatus</i>	Yes	Yes
<i>Boa constrictor</i>	Yes	Yes
<i>Boa dumerili</i>	No	Yes
<i>Boa imperator</i>	Yes	No
<i>Bombina orientalis</i>	Yes	Yes
<i>Borikenophis portoricensis</i>	No	Yes
<i>Caiman crocodilus</i>	Yes	Yes
<i>Calabaria reinhardtii</i>	No	Yes
<i>Calotes versicolor</i>	No	Yes
<i>Centrochelys sulcata</i>	Yes	Yes
<i>Chalcides ocellatus</i>	Yes	Yes

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<i>Chamaeleo calypttratus</i>	Yes	Yes
<i>Chelonoidis carbonarius</i>	Yes	Yes
<i>Chelonoidis denticulatus</i>	Yes	Yes
<i>Chilabothrus inornatus</i>	No	Yes
<i>Chilabothrus strigilatus</i>	No	Yes
<i>Chlamydosaurus kingii</i>	No	Yes
<i>Chondrodactylus bibronii</i>	No	Yes
<i>Chondrodactylus turneri</i>	No	Yes
<i>Chrysemys picta</i>	Yes	No
<i>Chrysopelea ornata</i>	No	Yes
<i>Cnemidophorus lemniscatus</i>	Yes	Yes
<i>Corallus hortulana</i>	Yes	Yes
<i>Crocodylus moreletii</i> *	Yes	Yes
<i>Crocodylus niloticus</i>	Yes	Yes
<i>Ctenosaura pectinata</i>	Yes	Yes
<i>Ctenosaura similis</i>	Yes	Yes
<i>Cyclura cornuta</i>	No	Yes
<i>Dendrelaphis pictus</i>	No	Yes
<i>Dracaena guianensis</i>	No	Yes
<i>Duttaphrynus melanostictus</i>	No	Yes
<i>Eleutherodactylus coqui</i>	Yes	Yes
<i>Eleutherodactylus planirostris</i>	Yes	Yes
<i>Emydura subglobosa</i>	Yes	No
<i>Epicrates cenchria</i> *	Yes	Yes
<i>Epicrates crassus</i>	No	Yes
<i>Erpeton tentaculatum</i>	No	Yes
<i>Eryx colubrinus</i>	No	Yes
<i>Eublepharis macularius</i> *	Yes	Yes
<i>Eunectes murinus</i>	No	Yes
<i>Eunectes notaeus</i>	No	Yes
<i>Eutropis multifasciata</i>	Yes	No
<i>Furcifer lateralis</i>	No	Yes
<i>Furcifer oustaleti</i>	Yes	Yes
<i>Furcifer pardalis</i>	Yes	Yes
<i>Gekko badenii</i>	Yes	Yes
<i>Gekko gecko</i>	Yes	Yes
<i>Gekko grossmanni</i>	Yes	Yes
<i>Gekko vittatus</i>	No	Yes
<i>Geochelone pardalis</i>	No	Yes
<i>Gerrhosaurus flavigularis</i>	Yes	No

<i>Gongylophis colubrinus</i>	No	Yes
<i>Gonocephalus grandis</i>	No	Yes
<i>Graptemys oculifera</i>	Yes	No
<i>Graptemys ouachitensis</i>	Yes	No
<i>Graptemys pseudogeographica</i>	Yes	No
<i>Hemidactylus frenatus</i>	Yes	Yes
<i>Hemidactylus garnotii</i>	Yes	Yes
<i>Hemidactylus mabouia</i>	Yes	Yes
<i>Hemidactylus parvimaculatus</i>	Yes	No
<i>Hemidactylus platyurus</i>	Yes	Yes
<i>Hemidactylus turcicus</i>	Yes	Yes
<i>Hemitheconyx caudicinctus</i>	No	Yes
<i>Hyla savignyi</i>	No	Yes
<i>Hypselotriton orientalis</i>	No	Yes
<i>Iguana iguana</i>	Yes	Yes
<i>Indotestudo elongata</i>	Yes	No
<i>Indotyphlops braminus</i>	Yes	Yes
<i>Kinixys homeana</i>	No	Yes
<i>Lacerta viridis</i>	Yes	No
<i>Langaha madagascariensis</i>	No	Yes
<i>Laudakia stellio</i>	No	Yes
<i>Leiocephalus carinatus</i>	Yes	Yes
<i>Leiocephalus schreibersii</i>	Yes	Yes
<i>Leiolepis belliana</i>	Yes	Yes
<i>Leiolepis rubritaeniata</i>	Yes	Yes
<i>Lepidodactylus lugubris*</i>	Yes	Yes
<i>Lygodactylus picturatus</i>	Yes	Yes
<i>Manouria emys*</i>	Yes	Yes
<i>Mauremys sinensis</i>	Yes	Yes
<i>Morelia spilota</i>	No	Yes
<i>Orthriophis taeniurus</i>	No	Yes
<i>Osteopilus septentrionalis</i>	Yes	Yes
<i>Pachytriton labiatus</i>	No	Yes
<i>Pelusios castaneus*</i>	No	Yes
<i>Pelusios sinuatus</i>	No	Yes
<i>Phelsuma grandis</i>	Yes	Yes
<i>Phelsuma laticauda*</i>	Yes	Yes
<i>Phrynosoma cornutum</i>	Yes	No
<i>Physignathus cocincinus</i>	No	Yes
<i>Pogona vitticeps</i>	Yes	Yes

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<i>Ptychozoon kuhli</i>	No	Yes
<i>Python bivittatus</i>	Yes	Yes
<i>Python breitensteini</i>	No	Yes
<i>Python curtus</i>	No	Yes
<i>Python regius</i>	Yes	Yes
<i>Python reticulatus</i>	No	Yes
<i>Python sebae</i>	No	Yes
<i>Ranoidea caerulea</i>	Yes	Yes
<i>Rhinella horribilis</i>	Yes	Yes
<i>Rhinella marina</i>	Yes	Yes
<i>Rhinoclemmys pulcherrima</i> *	Yes	Yes
<i>Salvator merianae</i>	Yes	Yes
<i>Salvator rufescens</i>	No	Yes
<i>Sphaerodactylus elegans</i>	Yes	Yes
<i>Staurotypus salvinii</i>	No	Yes
<i>Tarentola annularis</i>	Yes	Yes
<i>Tarentola mauritanica</i>	No	Yes
<i>Terrapene ornata</i>	Yes	No
<i>Testudo hermanni</i>	Yes	Yes
<i>Testudo horsfieldii</i>	Yes	Yes
<i>Tiliqua scincoides</i>	No	Yes
<i>Trachemys script</i>	Yes	Yes
<i>Trachemys venusta</i>	Yes	Yes
<i>Trachylepis quinquetaeniata</i>	Yes	Yes
<i>Tropidurus hispidus</i>	No	Yes
<i>Tupinambis teguixin</i>	No	Yes
<i>Typhlonectes natans</i>	Yes	Yes
<i>Uta stansburiana</i>	Yes	No
<i>Varanus albigularis</i>	No	Yes
<i>Varanus doreanus</i>	No	Yes
<i>Varanus exanthematicus</i> *	Yes	Yes
<i>Varanus indicus</i>	No	Yes
<i>Varanus jobiensis</i>	No	Yes
<i>Varanus niloticus</i>	Yes	Yes
<i>Varanus ornatus</i>	No	Yes
<i>Varanus panoptes</i>	No	Yes
<i>Varanus rudicollis</i>	No	Yes
<i>Varanus salvadorii</i>	No	Yes
<i>Varanus salvator</i>	Yes	Yes
<i>Xenopus laevis</i>	No	Yes

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<i>Zamenis scalaris</i>	Yes	No
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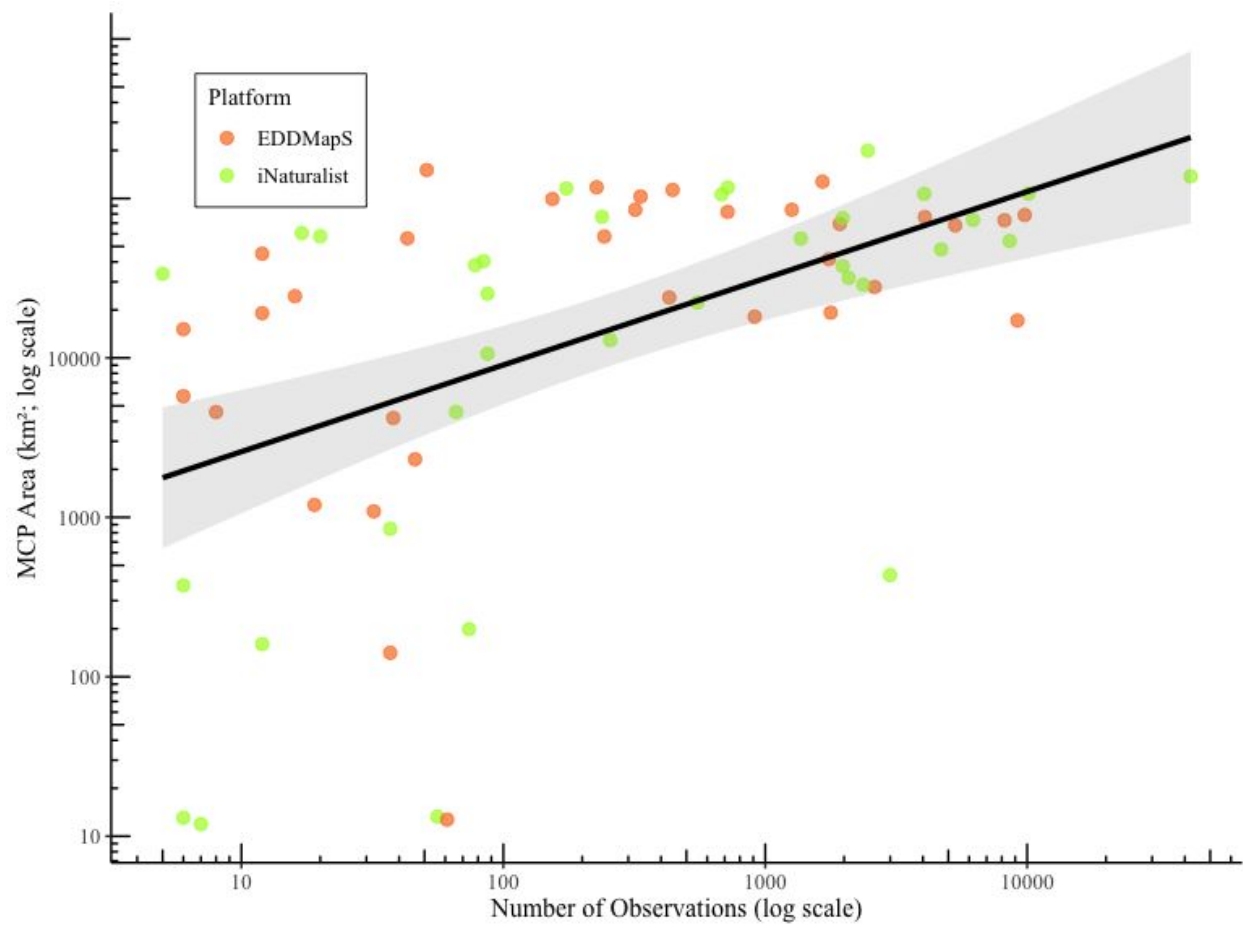


Figure S1: The positive relationship between the number of species observations and 95 minimum convex polygon (MCP) area. Each point represents a species-platform combination with green indicating iNaturalist and orange indicating EDDMapS. The regression line shows the overall fitted relationship with a shaded 95% confidence interval.